Lumiere Research Scholar Final Paper Template

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Acknowledgements
Thank you for the guidance of XXX mentor from YYY University in the development of this research paper.

***Abstract* - The abstract is primarily a summary of your paper. It is a shortly worded section of 100-250 words that outlines the research question and states the argument(s) presented in the paper. It briefly discusses the theoretical underpinnings and the methodology used to arrive at the argument as well (literature review, experiment, data analysis, etc.). In short, it is an overview of what is presented in the paper. Papers should include an abstract and a set of index terms. The Abstract and Index Terms here is in 10-point Times New Roman, fully justified and contained within one paragraph. Begin the Abstract with the word Abstract - in Times New Roman italic. The entire Abstract should be in bold and should not be indented. A standard dash is placed after the word "Abstract." Do not cite references or use abbreviations in the abstract. It should be approximately 125 - 175 words.**

Index Terms (optional) - ~Four alphabetically ordered key words or phrases, separated by commas (e.g., Camera-ready, FIE format, Preparation of papers, Two-column format).

I. Introduction

This section of the paper sets the base for the arguments and discussion presented. It moves from general description of the area of study to the specific gap in knowledge that you seek to address. The components of the introduction can often be divided into three sections:

a. Setting the context: This part should answer the following questions: why is your question relevant and/or interesting? Why does it matter? Every assertion should be referenced.
b. Research question, hypothesis, and aim: After explaining the context of the question, you should explicitly state the question. In addition, what do you seek to infer from this research? What is the inference that you seek to bring out through this paper, and why or how can it be justified?
c. Background/literature Review: This part seeks to offer context, background, and justification for the research. It includes a literature review, and the theoretical descriptions that underpins the argument in the paper. For some papers, this will the bulk of the content and should be its own section if data analysis and/or experimentation are not performed.

II. METHODOLOGY

The methodology section should explain what you did in your research so that anyone who reads it can replicate your exact process. Preciseness and detail are essential. You should describe your methods for choosing subjects, collecting data or measurements, and conducting analysis as specifically as possible. The methodology component provides an explanation of the experimental method or process for generating results, including materials. This may also apply to a proposed study or experiment; every step should be laid out so that readers can fully comprehend the steps taken to collect data that is unbiased and accurate. This section expounds on the sources of information, materials or the data used to test and evidence the hypothesis presented. It synthesizes information and argumentation for putting the sources in conversation with one another. If the paper is theory driven, this section helps in the development of the theory that underpins the paper. It expounds on the specific design and procedures implemented to test the hypothesis and arrive at an argument.

2.1 Participants

[Placeholder: Recruitment source, total N, age (M ± SD), gender breakdown, inclusion/exclusion criteria, and IRB/ethics statement.]

2.2 Research Design

This study employed a 2×2 between-subjects factorial design, with two independent variables: Video Format (short-form vs. long-form) and Total Viewing Duration (approximately 7 minutes vs. approximately 14 minutes). Each variable had two levels, and by combining the two variables together, there were four experimental conditions in total:

- Short-form video · Short duration (~7 minutes)
- Short-form video · Long duration (~14 minutes)
- Long-form video · Short duration (~7 minutes)
- Long-form video · Long duration (~14 minutes)

Participants were randomly assigned to one of the four conditions. Each participant was exposed to only one experimental treatment and was not informed of the existence of other conditions, in order to minimize demand characteristics.

The primary dependent variables in the study were four main parameters in the Sustained Attention to Response Task (SART): commission errors, omission errors, mean reaction time, and reaction time standard deviation. Because each participant took the SART test both before and after they watched the assigned videos, the actual dependent variables analyzed in the study were the change in SART scores, calculated as the scores after watching the videos minus the scores before that (post - pre). A positive change in scores indicates a decline and impairment in attentional state and performance, while a negative change indicates an improvement. By taking the change in scores instead of directly using scores after watching videos, the potential influence of individual baseline SART performance differences on the experiment result can be effectively diminished, thereby achieving fairer comparisons.

The secondary dependent variables were the three dimensions of self-reported mood from the questionnaire right after watching the videos: happiness, sadness, and energy level, each rated on a 1-5 Likert scale. These data served as a supplement to the main focus of this research, gaining insight into how emotional states indirectly influence attention performance, rather than the primary outcome measures.

The research's core logic is by simultaneously manipulating "video form" and "total exposed duration", to examine their respective effect on the attentional performance and subjective mood perception, even revealing a possible interaction effect between the two factors. This design managed to address multiple causal relationships within a single experiment.

2.3 Materials

Video Stimuli

The experiment prepared four groups of video stimulus materials, each corresponding to one experimental condition. The only difference between these video materials was the single video's length, with no preference or focus on any specific topic or video creator, in order to improve comparability and experimental control across conditions. Videos were randomly selected from YouTube's trending section and downloaded with the free Google Extension – Downr Media Downloader.

- Short-form video condition: The video stimuli consisted of multiple short clips downloaded from YouTube. Individual clips typically ranged from 15 to 60 seconds in length, covering entertainment, education, humor – categories typical for the short video recommendation algorithm – simulating the daily motion of pointlessly scrolling the screen and our precious attention being constantly fragmented.
- Long-form video condition: Two popular YouTube videos were selected as long-form stimulus – a 7-min animated introduction video of the famous Fermi paradox, and a 14-min entertaining video comparing GPT, Claude, and Gemini's ability to code the mobile game Subway Surfers and test their effect. Both videos blend the serious scientific or technological concept with an engaging and entertaining style. Each of the two videos served as a whole group and was played continuously without interruption, providing a coherent, attentive viewing experience.

Within each video format condition, stimuli were further divided into a short-duration version (~7 minutes) and a long-duration version (~14 minutes). The video content type was held constant within each format condition, with only total playback duration varying between duration levels. Specifically, for Group 1 and Group 2, both played short-form videos with each video lasting less than one minute. The only difference was that Group 1 consisted of 10 of these short videos, with a total duration of 7 minutes, and the numbers were 20 videos and 14 minutes for Group 2. On the other hand, for Groups 3 and 4, both groups played a single, long-form video, with one lasting 7 minutes and the other 14 minutes. All videos were presented in full-screen mode on the experimental platform and were set to be non-pausable and non-skippable, ensuring that all participants received the full experimental treatment for the designated video form and duration.

Questionnaire Materials

Baseline Questionnaire: Baseline information was gathered before the participants watch the videos, in order to collect covariates that might influence attentional performance, including: total sleep time of the previous night (in hours), type of primary activity in the preceding 30 minutes (e.g., studying, resting, exercising), current energy level (1-5 Likert scale self-rating), and current stress level (1-5 Likert scale self-rating).

Mood Questionnaire: Administered immediately after video viewing and before the post-test SART. The questionnaire included three dimensions, each rated on a 1–5 Likert scale (1 = not at all, 5 = very much):

- Happiness (Happy): "I feel happy and cheerful."
- Sadness (Sad): "I feel down and sad."
- Energy (Energetic): "I feel energetic and alert."

Demographics Questionnaire: Participants' basic information was gathered at the end of the experiment, right after the post-SART test, collecting information including age, gender, region, as well as self-reported social media usage habits (e.g., average daily time spent on short-form video platforms, primary usage type – passively scrolling / actively posting).

SART Task: The Sustained Attention to Response Task (SART) used in this study was based on the classic paradigm developed by Robertson et al. (1997), adapted for online administration. The task ran entirely in a web browser with no software installation required; participants responded using the spacebar on a standard keyboard.

SART Task: The Sustained Attention to Response Task (SART) was designed and programmed based on the paradigm established in the original paper by Robertson et al. (1997), and was adapted for online testing. The task ran entirely on a website with no software installation required; participants responded using the normal spacebar on their keyboard.

Task parameters were as follows, all of which correspond to the original and standard test:

- Digit display duration: Each digit was presented on the screen and lasted for 250 milliseconds (digitMs = 250 ms).

- Mask duration: Each digit was followed by a masking stimulus displayed for 900 milliseconds (maskMs = 900 ms), yielding a total stimulus onset asynchrony (SOA) of 1,150 ms per trial, in order to avoid any visual afterimage of the preceding digit interfering with the current trial.
- Total number of trials: 225 trials in the main test section (mainTrials = 225); an additional 9 slow-paced practice trials (practiceTrials = 9) were administered during the practice phase only, with immediate correctness feedback provided right after each response. Practice data was not recorded.
- Font sizes: Digits were presented in five randomly varying font sizes — 48, 72, 94, 100, and 120 pixels — to introduce visual uncertainty and prevent participants from anticipating target stimuli based on size alone.
- Target stimulus: Digit 3 was set to be the No-Go target. Participants were instructed to withhold their response (i.e., not press the spacebar) when digit 3 appeared, and to press the spacebar as quickly as possible for all other digits (1, 2, 4, 5, 6, 7, 8, 9).
- Platform: The experiment was hosted on GitHub Pages and was deployed online, compatible with most browsers. No account login was required. Response data were automatically connected and written to a Google Sheets database via Google Apps Script every time a participant completed the experiment.

2.4 Procedure

The experiment was conducted online throughout the whole process. Participants accessed the experimental website via a shared public link deployed on GitHub, requiring no laboratory visit or software installation.

Upon accessing the website, the participants first read the consent form, which strictly followed the standard IRB format and described the general experiment information, including purpose, content, and duration. The participants were also informed of the fact that they have the right to withdraw at any time. To protect the participants' anonymity and privacy, the experiment did not collect any personally identifiable information, and the data was marked by random IDs instead of participants' names. After providing their consent via the "I agree" checkbox, participants completed the brief baseline questionnaire reporting their basic attentional states, such as last night's sleep duration, primary activity type (study / relaxing), and evaluated their current energy and stress levels on a 1-5 scale.

Participants then receive written instructions for the SART test, followed by a practice section of 9 slow-paced trials with immediate feedback on their responses. This practice design aimed to help the participants better understand the rules and avoid their different perceptions of the test acting as a confounding variable. Once they finished the practice, the participants proceeded to the formal pre-SART test, finishing 225 trials without feedback on the correctness. The result from the pre-SART test served as each individual's performance baseline.

Following the pre-SART test, the participants were automatically and randomly assigned to one of the four experimental groups and watched the corresponding videos in fullscreen. The videos could not be paused, replayed, or skipped, ensuring that all participants received the full-length stimuli. Immediately after the video ended, participants completed a subjective mood questionnaire where they rated their current happiness, drowsiness, and energy level on a 1-5 Likert scale. Then they proceeded to the post-SART test under the same parameters as the pre-SART test. The post-SART test could reveal the effect of the video stimuli, and its difference with the post-SART test was recorded and finally analyzed.

The experiment concluded with a demographic questionnaire collecting basic information such as age, gender, and location. Participants would then be shown a brief result page displaying a side-by-side comparison of their pre- and post-SART test performance. The total session duration was approximately 20 to 30 minutes, depending on the assigned video group.

2.5 Measures

All measures were derived from participants' keypress behavior during the SART and comprised the following four indices:

（1）**Commission Errors** Defined as the number of trials on which the participants incorrectly pressed the space button in response to the target digit 3. This index reflects failure of response inhibition – participants were unable to successfully suppress their temporal habit of continuously pressing the key bar. Within the SART paradigm, commission errors are considered to be the core measure of sustained attentional state: a higher commission error rate indicates more frequent

attentional lapses, during which the participants partially lost their ability to inhibits certain behaviors and instead respond automatically.

（2）**Omission Errors** Defined as the number of trials where the participants did not press the spacebar correctly inresponse to a non-target digit (i.e., any digit other than 3). This measure indicates lack of response preparedness or attentional disengagement. When the overall response alertness decreases, omission errors tend to increase correspondingly.

（3）**Mean Reaction Time (Mean RT)** Defined as the average response time of all correctly responded non-target trials (measured in milliseconds). Only correct responses were included in the calculator. Incorrectly pressing the spacebar on the target trails were excluded to avoid some strategically slowed responses of the participants contaminating the calculation. Longer mean RT indicates decreased response alertness and weakened attentional state.

（4）**Reaction Time Standard Deviation (RT SD)** Defined as the response time's level of fluctuation among the correctly responded trials, measured in standard deviations (in ms). RT SD serves as the indicator of the attentional stability, since when one is distracted, his response time will fluctuate significantly. Sometimes even if there is no notable change in SART scores when comparing the pre-SART test and post-SART test, RT SD can increase significantly. Research has shown that RT SD is especially sensitive to detect mind-wandering and attentional instability.

Change Score Calculation

All four measures were separately calculated in the pre-SART test and post-SART test, recording four test values in each of the two groups. The actual dependent variables used in the final analysis were change scores, calculated as:

$$\Delta \text{ index} = \text{post-SART test value (post)} - \text{pre-SART test value (pre)}$$

- **Δ commission errors** > 0: decline in response inhibition (more errors on digit 3 after video viewing)
- **Δ omission errors** > 0: decline in overall response readiness
- **Δ mean RT** > 0: slower cognitive processing speed (responses became slower)
- **Δ RT SD** > 0: decline in attentional stability (greater trial-to-trial variability in response speed)

The use of change scores rather than post-test scores alone was intended to statistically control the individual differences in baseline performance – for example, participants who are naturally slower or worse responders or who have inherently higher error rates – thereby enabling causal attribution of observed differences to the experimental treatment of video viewing itself.

Comparing the two groups' scores and analyzing the difference rather than post-SART test alone was intended to control the exclude the effect of individual differences in baseline performance – for example, participants who naturally have slower response time probably have worse performance in the SART tests – thereby effectively ruling out the confounding variable and revealing the potential causal relationship between the experimental treatment and the attentional state.

2.6 Data Analysis

[Placeholder: Mixed ANOVA (2 Video Format x 2 Viewing Duration) on each change-score DV (commission errors, omission errors, mean RT, RT SD); main effects and interaction term reported; effect sizes (partial eta-squared); post-hoc comparisons if interaction significant; all analyses conducted in R or JASP; alpha = .05.]

<u>If the paper is not experimental based, this section could be added to the introduction and be replaced with a comprehensive literature review section that synthesizes information and argumentation of the sources in conversation with one another. There is no need to justify the process of doing a literature review (i.e. stating how sources were vetted or targeting key words)</u>

III. RESULTS

These are the "answers" that your research produced. This is not a presentation of raw data, but a presentation of the numbers or facts determined from the analysis. If you are using a data-driven hypothesis, this is where you state whether you accept or reject that hypothesis given the information presented. Properly formatted tables and charts can make this section brief. It is not always necessary to repeat in paragraphs what you show in figures and charts. Text should lead the reader to the chart or table and emphasize key findings rather than repeat what can be visualized. <u>This section should be purely descriptive and should include no judgements or opinions.</u>

IV. Figures, Tables and Equations

All figures and tables should fit either one or two-column width, 3.4" or 7" wide respectively. It is suggested that you use one-column whenever possible. If your table or figure will not fit into one-column, then insert a continuous section break before and after the table or figure, as described above and define it as one-column. To make the paper read easier you may want to position any table or figure that requires one-column either at the bottom of the page or the top of a new page.

Points	Place of Text	Type Styles
8	Table number	Roman numerals
8	Table text	
8	Figure and Table Headings	UPPERCASE
8	Figure and Table Captions	Small Caps
8	Footnote	
8	Reference list	
10	Footer	**Bold**
10	Abstract	**Bold**
10	Index Terms	
10	Section Titles	**Small Caps, Bold**
10	Main Text and Equations	
10	Subheadings	*Italic*, Left justified
11	Authors' names	
24	Title	Title Case

TABLE I
Point Sizes and Type Styles

BIRTHS AND FERTILITY RATES: UNITED STATES, FINAL 2005-2010, PRELIMINARY 2011, AND PROVISIONAL FOR THE 12 MONTHS ENDING JUNE 2012

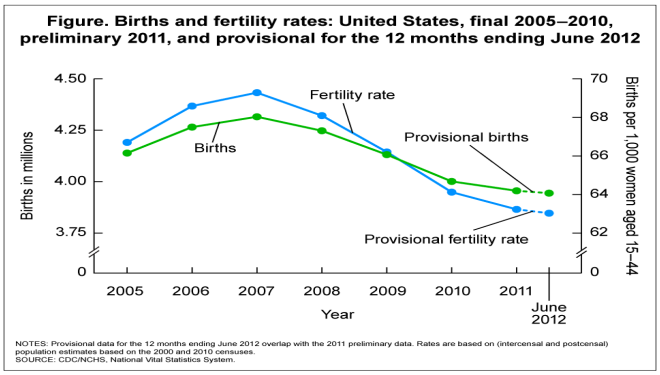

FIGURE 1

Do not abbreviate "Table"; use Roman numerals to number tables. Use the following format guidelines for figures and tables:

- **Figure and table headings**: 8 point, Times New Roman UPPERCASE, centered. Place below the figure and above the Table, (this style is defined under the style menu of this document as "Figure Heading")
- Leave one blank line above and below each Table or Figure.

- **Figure and table captions**: 8 point, Times New Roman, SMALL CAPS, centered. Place below the figure or table headings (this style is defined under the style menu of this document as "Figure Caption")
- **Table text**: 8 point, Times New Roman, (this style is defined under the style menu of this document as "Table text")

IV. DISCUSSION

In this section, you should avoid repeating what you showed in the "Results" section. Essentially, this section answers the question "What do these findings mean?". This is where you may offer analysis, principles, relationships, generalizations, etc. Develop paragraphs based on critical themes and trends revealed in the findings. Identify points that lack correlation or offer exceptions. Show how your research agrees or disagrees with similar or prior studies. This section connects the results to the hypothesis proposed in the introduction. This section provides the scope to introduce your subjective opinions and your interpretations of the results. This section should also include any drawbacks or limitations that your study presented. This is the "so what?" section.

V. CONCLUSION

A good conclusion should restate your answer to your research question, hypothesis, or primary claim based on your findings. It should also make recommendations for further studies or changes that should be put in practice. It should only be one to two paragraphs long. It should summarize the results and the discussion and expound on the implications of the discussion presented in the paper. The conclusion should elaborate on the significance, relevance, and generality of the argument presented. Often, conclusions close with future steps; what should or can be done to further advance the state of the subject and field.

REFERENCES

Place references in a separate References section; do not footnote references. Number the references sequentially by order of appearance, not alphabetically. Refer simply to the reference number, as [3] or [5]-[8]. Do not use "Ref. [3]" or "reference [3]" except at the beginning of sentence: "Reference [3] shows…." Provide up to three authors' names; replace the others by "*et al*." Do not put figures or anything else except for author information after the references.

- **Reference text**: 8 point, Times New Roman, full justified, no space between the references (this style is defined under the style menu of this document as "References")
- Use box numbers with square brackets [] within text. Do not use superscripts or subscripts. Do not use () for references, since these are used to refer to equations.
- The style of citations differs based on the fields of research, and thus it is best to consult your mentors about what to use in their field.

Use the following as examples:

[1] G. O. Young, "Synthetic structure of industrial plastics (Book style with paper title and editor)," in *Plastics*, 2nd ed. vol. 3, J. Peters, Ed. New York: McGraw-Hill, 1964, pp. 15–64.

[2] W.-K. Chen, *Linear Networks and Systems* (Book style). Belmont, CA: Wadsworth, 1993, pp. 123–135.

[3] H. Poor, *An Introduction to Signal Detection and Estimation*. New York: Springer-Verlag, 1985, ch. 4.

[4] B. Smith, "An approach to graphs of linear forms (Unpublished work style)," unpublished.

[5] E. H. Miller, "A note on reflector arrays (Periodical style—Accepted for publication)," *IEEE Trans. Antennas Propagat.*, to be published.

[6] J. Wang, "Fundamentals of erbium-doped fiber amplifiers arrays (Periodical style—Submitted for publication)," *IEEE J. Quantum Electron.*, submitted for publication.